HINT

It is essential that you don't try to learn these equations. In an exam, you could be given one of a wide range of possible hydrocarbons to start with. You need to understand what is happening, so that you can be adaptable.

The molecule might have split quite differently. All sorts of reactions are happening in a catalytic cracker. Two other possibilities might be:

$$C_{13}H_{28}(l) \rightarrow 2C_2H_4(g) + C_9H_{20}(l)$$

$$C_{13}H_{28}(l) \rightarrow 2C_2H_4(g) + C_3H_6(g) + C_6H_{14}(l)$$

Some reactions might even produce a small percentage of free hydrogen. For example:

$$C_{13}H_{28}(l) \rightarrow 2C_2H_4(g) + C_3H_6(g) + C_6H_{12}(l) + H_2(g)$$

In this case, all the hydrocarbons formed will have double bonds. They are all alkenes.

There will also be very many other ways in which this particular molecule might have split up. Also, remember that the fraction being cracked contains a complex mixture of hydrocarbons, not just one. At the end of the cracking process you will have an equally complex mixture of smaller hydrocarbons, both alkanes and alkenes. This mixture will have to go through a lot of further processing (including further fractional distillation) to separate everything out into pure compounds.

EXTENSION WORK

In industry, catalytic cracking is done by passing hydrocarbons through a bed of zeolite (aluminosilicate) catalyst at about 500 °C and moderate pressures. Apart from breaking down large straight-chain molecules into smaller ones, isomerisation can occur during the process to produce branched-chain alkanes and alkanes with ring structures. The presence of these isomers in a fuel helps to increase the *octane number* of a fuel, which is a measure of the tendency of a fuel to not undergo auto-ignition in an engine. *Auto-ignition* is when fuel burns spontaneously out of control. It leads to wear in the engine and wastage of petrol. The higher the octane number, the lower the tendency for a fuel to undergo auto-ignition.

CHAPTER QUESTIONS

SKILLS CRITICAL THINKING (5)

1 Crude oil is a mixture of hydrocarbons.

a State which two elements are present in the compounds in crude oil.

(7) b Crude oil is separated into fractions by heating and passing the vapour into a fractionating column. Explain how the fractions separate in the column.

(6) c Two of the fractions are gasoline and diesel. State one use of each.

d Name two fractions formed in the fractional distillation of crude oil, other than gasoline and diesel.

e Describe the differences between gasoline and diesel. In your answer you should refer to the average size of the molecules in the two liquids, the colour of the two liquids and the viscosities of the two liquids.

(7) f One of the hydrocarbons found in the diesel fraction is $C_{18}H_{38}$. Suggest the formula of a hydrocarbon that could be found in the gasoline fraction.

SKILLS PROBLEM SOLVING (9)

g i Write a balanced chemical equation for the complete combustion of butane, C_4H_{10}.

SKILLS REASONING (6)

ii Incomplete combustion of hydrocarbons produces carbon monoxide. Explain why carbon monoxide is harmful to humans.

SKILLS REASONING



2 Explain why:

a burning a fuel containing sulfur as an impurity causes acid rain

b petrol engines produce oxides of nitrogen

c the presence of too much sulfur dioxide, SO_2, in the atmosphere has a damaging effect on many buildings made of marble, a form of calcium carbonate, $CaCO_3$.

SKILLS CRITICAL THINKING

3 Cracking is a process that splits larger hydrocarbons into smaller ones.

a Give two reasons why an oil company might want to crack a hydrocarbon.

b Give the conditions under which cracking is carried out.

SKILLS PROBLEM SOLVING



c A molecule of the hydrocarbon $C_{11}H_{24}$ was cracked to give two molecules of ethene, C_2H_4, and one other molecule. Write a balanced chemical equation for the reaction which took place. (You can omit the state symbols from your equation.)

d Write an equation for an alternative cracking reaction involving the same hydrocarbon, $C_{11}H_{24}$.

SKILLS CREATIVITY

4 Imagine a world in which fossil fuels such as coal, natural gas and oil had never formed. This would have effects other than the immediately obvious ones. For example, the iron and steel industry depends on coke made from coal, although in the past (on a much smaller scale) it used charcoal made from wood. Choose one aspect of the modern world and explain, in no more than 300 words, how it would be different in the absence of fossil fuels. You could choose from transport, use of materials, landscape, disease prevention, and power generation, for example, but you don't have to restrict yourself to this list.

Try not to be too simplistic about this, human beings are inventive! For example, it is possible to obtain organic chemicals from alcohol (from fermented sugar) and fuels from plant oils.